Differential Equations (Ext 1 & 2)

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Booklet repository:

https://github.com/vuhung16au/math-olympiad-ml/

"Talent is important, but how one develops and nurtures it is even more so."

Terence Tao

Contents

1 Introduction

1.1 Project Overview

Differential equations sit at an important meeting point between algebra, calculus, graphs, and mathematical modelling. A differential equation does not ask only for a number: it asks for a whole function whose rate of change satisfies a given rule. This makes the topic especially powerful, because many real situations are naturally described by rates. Population growth, cooling, motion, and the spread of a quantity through time can all be framed in this language.

This booklet is written as a syllabus-aligned guide for HSC Mathematics Extension 1 students, with clear Extension 2 connections and a small amount of carefully signposted enrichment for students who want to move further toward university mathematics, engineering, or mathematical modelling. The core emphasis is on first-order differential equations, slope fields, separable forms, and logistic-style models.

1.2 Where This Topic Sits in the NESA Pathway

In the NSW Stage 6 pathway used for the current HSC cycle, differential equations are part of the **Mathematics Extension 1** calculus strand. They are examinable there as a core topic alongside applications of calculus. Students studying **Mathematics Extension 2** meet the ideas again through concurrent study, especially when rates of change are used in mechanics and higher modelling contexts.

That means the main body of this booklet treats differential equations as **Extension 1 core material**. When an idea goes beyond that level, it is labelled as enrichment so students can clearly separate examinable HSC content from helpful mathematical extension.

1.3 Who This Booklet Is For

This resource is intended for:

- Extension 1 students who want a focused and readable guide to differential equations,
- Extension 2 students who want to connect calculus ideas more strongly with modelling and mechanics,
- motivated learners who want a first look at how school calculus grows into university mathematics and engineering.

Teachers and tutors can also use the structure as a teaching sequence: start with meaning, move to slope fields, then solve algebraically, and finally interpret solutions in context.

1.4 How to Use This Booklet

Differential equations are easiest to learn when the graphical and algebraic viewpoints are developed together. A good rhythm is:

1. read the short concept summary,
2. sketch what the slope field or long-run behavior suggests,
3. solve the equation carefully,
4. apply any initial condition,

5. interpret what the solution means in the original context.

Try to resist the temptation to treat every question as pure symbol manipulation. In this topic, the best answers usually connect three viewpoints at once: the equation, the graph, and the interpretation.

1.5 Prerequisite Knowledge

Before working through this booklet, students should be comfortable with:

- derivatives as rates of change,
- anti-derivatives and basic integration,
- separation of variables in simple algebraic settings,
- exponential and logarithmic functions,
- graph sketching and the interpretation of gradients.

1.6 Core Syllabus vs Enrichment

Core HSC focus:

- recognizing a differential equation,
- interpreting a slope field,
- solving simple first-order equations,
- working with equations of the forms $y' = f(x)$, $y' = g(y)$, and $y' = f(x)g(y)$,
- using initial conditions,
- interpreting logistic-style growth models.

Enrichment focus:

- more formal stability language,
- broader modelling ideas,
- links to mechanics and engineering,
- first-order linear equations beyond the main HSC treatment.

2 Fundamentals Review

This short review gathers the basic language, notation, and standard forms that appear throughout the booklet.

2.1 Differential equations and solutions

A differential equation is an equation involving an unknown function and one or more of its derivatives. In this booklet, the focus is on first-order differential equations, where the highest derivative is y'.

- A **general solution** contains an arbitrary constant such as C.
- A **particular solution** is found after applying a condition such as $y(0) = 3$.
- An **initial condition** selects one curve from a whole family of solutions.

$$\frac{dy}{dx} = 2x \qquad \Longrightarrow \qquad y = x^2 + C.$$

2.2 Direction fields and equilibrium solutions

For an equation of the form $\frac{dy}{dx} = F(x, y)$, a slope field places a short line segment at each point (x, y) with slope $F(x, y)$. This gives a visual picture of possible solution curves.

- If the slope is positive, solution curves rise from left to right.
- If the slope is negative, solution curves fall from left to right.
- If the slope is zero along a line $y = k$, then $y = k$ is an equilibrium solution.

2.3 Separable structure

The separable forms most relevant to HSC Extension 1 are:

$$\frac{dy}{dx} = f(x), \qquad \frac{dy}{dx} = g(y), \qquad \frac{dy}{dx} = f(x)g(y).$$

When variables can be separated, the usual workflow is:

1. Rearrange so all y terms are with dy and all x terms are with dx.
2. Integrate both sides.
3. Apply any initial condition.
4. Interpret the result in context.

2.4 Logistic-style models

The logistic equation is commonly written as

$$\frac{dy}{dx} = ky(M - y),$$

or equivalently

$$\frac{dy}{dx} = ry\left(1 - \frac{y}{M}\right),$$

where M is the carrying capacity.

- $y = 0$ and $y = M$ are equilibrium values.
- If $0 < y < M$, growth is positive.
- If $y > M$, the model predicts decay back toward the carrying capacity.

3 Differential Equations

3.1 Concept Overview

A differential equation is an equation involving an unknown function and one or more of its derivatives. In the HSC setting, we mainly work with first-order equations, where the derivative y' describes the slope of the solution curve at each point.

The key shift in thinking is this: instead of asking for a single value, we are looking for a function whose rate of change matches the given rule. Often there is not just one answer, but a family of answers.

3.2 Key Forms

$$\frac{dy}{dx} = F(x, y), \qquad \frac{dy}{dx} = f(x), \qquad \frac{dy}{dx} = g(y), \qquad \frac{dy}{dx} = f(x)g(y).$$

$$\text{Initial condition example: } y(0) = 2.$$

3.3 Interpretation Notes

- A general solution usually contains a constant of integration.
- An initial condition turns a family of curves into one particular solution.
- The derivative tells us the local behavior of the solution: whether it rises, falls, or stays flat.

3.4 Typical Mistakes

- Forgetting the constant of integration.
- Applying an initial condition too early or incorrectly.
- Treating the derivative as a disconnected algebraic symbol rather than a rate or slope.

Enrichment

In higher mathematics, one also studies whether a solution exists, whether it is unique, and how sensitive it is to the starting condition. At HSC level, the main emphasis is on recognizing the structure, solving standard forms, and interpreting the result clearly.

4 Slope Fields

4.1 Concept Overview

A slope field, also called a direction field, is a visual way to display the differential equation $y' = F(x, y)$. At each point in the plane, a short line segment is drawn with slope $F(x, y)$. Solution curves then thread through those segments.

This gives a qualitative picture of the equation before any algebraic solving begins. It is especially useful for identifying equilibrium solutions and predicting long-run behavior.

4.2 Key Ideas

- If the segment is horizontal, then the slope is zero.
- If the segment tilts upward, then the solution increases locally.
- If the segment tilts downward, then the solution decreases locally.
- A constant solution $y = k$ is an equilibrium if the slope is zero there.

4.3 Interpretation Notes

- A particular solution must follow the local tangent directions shown by the field.
- If nearby segments steer curves toward a horizontal line, that line behaves like a long-run level.
- A full answer may require both a sketch and a written description of behavior as $x \to \infty$ or $x \to -\infty$.

4.4 Typical Mistakes

- Drawing curves that cut across the local directions.
- Ignoring the given starting point.
- Describing long-run behavior only on the graph and not in words.

Enrichment

Slope fields are an early example of qualitative analysis. Even when an equation is hard to solve exactly, the field can still reveal equilibrium levels, monotonic behavior, and asymptotic trends.

5 Separable Differential Equations

5.1 Concept Overview

Many HSC differential equations are separable, meaning the y terms and x terms can be placed on opposite sides before integrating. This is the main algebraic technique students are expected to master.

5.2 Canonical Forms

$$\frac{dy}{dx} = f(x), \qquad \frac{dy}{dx} = g(y), \qquad \frac{dy}{dx} = f(x)g(y).$$

For a separable equation, we aim for a structure like

$$A(y)\,dy = B(x)\,dx.$$

5.3 Interpretation Notes

- The form $y' = f(x)$ means the slope depends only on x.
- The form $y' = g(y)$ means the slope depends only on the current value of y.
- The form $y' = f(x)g(y)$ mixes both influences and is often the richest case.

5.4 Typical Mistakes

- Separating variables incorrectly.
- Dropping absolute values when integrating logarithmic forms.
- Forgetting to substitute the initial condition back into the integrated solution.

Enrichment

At university level, students later meet equations that are not separable and need different methods. Learning to recognize separable structure now builds the pattern-recognition needed for those later techniques.

6 Autonomous Equations and the Logistic Model

6.1 Concept Overview

An autonomous differential equation has the form $y' = g(y)$, so the slope depends only on the current value of y. These equations often have visible equilibrium values and clear long-run behavior.

The logistic equation is a central example because it combines growth with a limiting effect. It models a quantity that grows when small but slows as it approaches a carrying capacity.

6.2 Key Forms

$$\frac{dy}{dx} = g(y)$$

$$\frac{dy}{dx} = ky(M-y) \qquad \text{or} \qquad \frac{dy}{dx} = ry\left(1 - \frac{y}{M}\right).$$

6.3 Interpretation Notes

- Equilibrium values occur where $g(y) = 0$.
- In the logistic model, $y = 0$ and $y = M$ are equilibrium values.
- If $0 < y < M$, growth is positive.
- If $y > M$, the rate is negative and the model pushes back toward M.

6.4 Typical Mistakes

- Treating the carrying capacity as just another constant without explaining its meaning.
- Forgetting to check what happens above and below each equilibrium.
- Solving correctly but failing to interpret whether the answer is realistic in context.

Enrichment

The language of stable and unstable equilibria becomes very important in engineering, dynamical systems, and mathematical biology. The logistic equation is often a student's first real encounter with this kind of thinking.

7 Applications of Differential Equations

7.1 Concept Overview

Differential equations are useful because many real-world relationships are naturally phrased in terms of rates. Once a rate law is written down, calculus turns the model into a function that can be analyzed and interpreted.

7.2 Common Contexts

- growth and decay,
- population models,
- cooling or approach-to-limit models,
- motion and velocity-style rate models,
- simple contextual problems from science, economics, or engineering.

7.3 Interpretation Notes

- Always define what each variable represents.
- State what constants mean in words, not only symbols.
- Check whether the units and long-run behavior make sense.
- A mathematically valid answer may still be a poor model if the interpretation is unrealistic.

7.4 Typical Mistakes

- Translating a word statement into the wrong differential equation.
- Ignoring the meaning of the initial condition.
- Giving a symbolic answer without any contextual interpretation.

Enrichment

In Extension 2 mechanics and later university study, many modelling problems are still driven by the same habit: express a rate law clearly, solve the resulting equation carefully, and then return to the physical meaning of the answer.

8 Problems

This section is scaffolded only for now. The tier structure is in place so problems and solutions can be added in a later iteration without changing the overall booklet layout.

8.1 Warm-Up

Problems

- Placeholder: Warm-Up Problem 1
- Placeholder: Warm-Up Problem 2
- Placeholder: Warm-Up Problem 3

Solutions

- Placeholder: Warm-Up Solution 1
- Placeholder: Warm-Up Solution 2
- Placeholder: Warm-Up Solution 3

8.2 Medium

Problems

- Placeholder: Medium Problem 1
- Placeholder: Medium Problem 2
- Placeholder: Medium Problem 3

Solutions

- Placeholder: Medium Solution 1
- Placeholder: Medium Solution 2
- Placeholder: Medium Solution 3

8.3 Advanced

Problems

- Placeholder: Advanced Problem 1
- Placeholder: Advanced Problem 2
- Placeholder: Advanced Problem 3

Solutions

- Placeholder: Advanced Solution 1
- Placeholder: Advanced Solution 2
- Placeholder: Advanced Solution 3

9 Appendices

9.1 Appendix A: Differential Equations Quick Reference

- **General first-order form:** $\dfrac{dy}{dx} = F(x, y)$.
- **Initial condition:** a value such as $y(x_0) = y_0$ used to pick out one particular solution.
- **Separable forms:**

$$\frac{dy}{dx} = f(x), \qquad \frac{dy}{dx} = g(y), \qquad \frac{dy}{dx} = f(x)g(y).$$

- **Core integration idea:** if variables can be separated, rewrite the equation in the form

$$A(y)\,dy = B(x)\,dx$$

  and integrate both sides.
- **Equilibrium solution:** a constant solution $y = k$ for which $\dfrac{dy}{dx} = 0$.
- **Slope field reading rule:** each small segment shows the local tangent direction of a solution curve.
- **Exponential-style model:**

$$\frac{dy}{dx} = ky \qquad \Longrightarrow \qquad y = Ae^{kx}.$$

- **Logistic model:**

$$\frac{dy}{dx} = ry\left(1 - \frac{y}{M}\right),$$

  where M is the carrying capacity and the equilibrium values are $y = 0$ and $y = M$.
- **Interpretation checklist:** identify units, define constants clearly, apply the initial condition, and state what the long-run behavior means in context.

9.2 Appendix B: Reading Slope Fields

When reading a slope field, focus on the pattern before trying to draw a full solution curve.

1. **Look for flat segments.** Wherever the slope is zero, solution curves have horizontal tangents. A whole horizontal band of zero slopes often signals an equilibrium solution.
2. **Check sign regions.** If most segments in a region tilt upward, then solutions increase there. If they tilt downward, solutions decrease there.
3. **Track steepness.** Steeper segments indicate larger magnitude of derivative, so the solution changes more rapidly.
4. **Use the initial point carefully.** A particular solution must pass through the given starting point and then follow the local direction pattern.
5. **Think about end behavior.** If nearby segments steer a curve toward a horizontal line, the solution may approach an equilibrium as $x \to \infty$ or $x \to -\infty$.

Typical observations

- If slopes depend only on y, then rows often repeat horizontally.
- If slopes depend only on x, then columns often repeat vertically.
- If the field changes sign above and below an equilibrium, that equilibrium may behave like an attracting or repelling level.

Common mistakes

- Drawing a curve that cuts across the local segment directions.
- Forgetting that a slope field shows tangents, not disconnected graph pieces.
- Giving only a sketch when the question also asks for a verbal description of the long-run behavior.

9.3 Appendix C: Modelling Workflow

Many differential equations begin with language such as "the rate of change is proportional to..." or "the rate depends on the current amount." A clean modelling workflow helps turn those words into mathematics.

1. **Define the variable.** State what y represents and what the independent variable is. For example, y could be a population and x could be time in years.
2. **Translate the rate statement.** If the growth rate is proportional to the amount present, write
$$\frac{dy}{dx} = ky.$$
If the rate depends on both the amount present and remaining capacity, write a logistic-style equation.
3. **Solve the differential equation.** Use separation of variables or a known standard form.
4. **Apply the condition.** Use initial data such as $y(0) = y_0$ to determine the constant.
5. **Interpret constants.** Explain what k, r, or M mean in the situation.
6. **Check the answer against the story.** Ask whether the function stays positive when it should, whether units are sensible, and whether the long-run behavior is realistic.

Useful sentence patterns

- "Rate proportional to the current amount" usually leads to $\dfrac{dy}{dx} = ky$.
- "Rate proportional to the difference from a limiting value" often leads to $\dfrac{dy}{dx} = k(M - y)$.
- "Growth slows as the population approaches a maximum level" suggests a logistic model.

9.4 Appendix D: Enrichment and Where This Goes Next

This appendix is intentionally beyond the strict core of the booklet. It is included for students who want a clearer bridge to university mathematics, engineering, and Extension 2 style applications.

First-order linear equations

At a higher level, many important models are written in the form

$$\frac{dy}{dx} + P(x)y = Q(x).$$

These are called first-order linear differential equations. They are usually solved by an integrating factor rather than by direct separation. This method is valuable in university mathematics, physics, and engineering, but it is not the central HSC focus here.

Stability language

When an equilibrium solution is nearby, we can ask what happens to solutions that start close to it.

- An equilibrium is often called **stable** if nearby solutions move toward it.
- It is often called **unstable** if nearby solutions move away from it.

This language is common in differential equations, control theory, and dynamical systems.

Links to mechanics

Extension 2 mechanics repeatedly uses rates of change. Velocity is the rate of change of position, and acceleration is the rate of change of velocity. As a result, many mechanics questions naturally become differential equations, especially when forces depend on displacement, velocity, or both.

Where this goes next

Students moving further into mathematics, physics, economics, or engineering will later meet:

- linear differential equations,
- systems of differential equations,
- numerical methods,
- oscillation and stability,
- mathematical modelling of real systems.

The HSC treatment is only the beginning, but it introduces the most important habit early: connect algebra, graphs, and interpretation instead of treating them as separate skills.

10 Conclusion

Differential equations are one of the clearest examples of how calculus becomes a modelling language. The topic asks students to do more than differentiate and integrate: it asks them to connect rate, graph, formula, and interpretation in a single chain of reasoning. That combination is exactly why the topic is so useful later in Extension 2, mechanics, engineering, and university mathematics.

As you work through this booklet, aim to build two habits together: solve carefully, and explain what the solution means. Strong HSC answers usually do both. Once the problem bank is filled in, this booklet will become a natural place to practice that full cycle repeatedly.

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